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RESEARCH ARTICLE

Doubly Efficient Fuzzy Private Set Intersection for High-Dimensional Data With Cosine Similarity

HYUNJUNG SON[1], SEUNGHUN PAIK[1], (Graduate Student Member, IEEE), YUNKI KIM[1], SUNPILL KIM[1], (Graduate Student Member, IEEE), HEEWON CHUNG[2], AND JAE HONG SEO[1]

[1]Department of Mathematics, Hanyang University, Seoul 04763, Republic of Korea
[2]Department of Software Engineering, Jeonbuk National University, Jeonju 54896, Republic of Korea

Corresponding author: Jae Hong Seo (jaehongseo@hanyang.ac.kr)

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ABSTRACT Fuzzy private set intersection (fuzzy PSI) is a cryptographic protocol that enables privacy-preserving similarity matching, where a client securely learns which of its items are sufficiently close to some item in the service provider's dataset. This functionality is essential for various real-world applications, including biometric authentication, information retrieval, or recommendation systems. However, existing fuzzy PSI protocols suffer from two major barriers to deployment. First, many cannot handle high-dimensional feature vectors, e.g., from 128 to 512, in practical applications because of their exponential computation/communication overheads in dimension. Furthermore, existing protocols supporting L_2 distance cannot accommodate cosine similarity. We found that their distributional assumptions on the data enforce overly strict similarity matching thresholds for the application, leading to prohibitively large false negatives. In this paper, we present FPHC, the first fuzzy PSI protocol that efficiently handles high-dimensional data with cosine similarity. FPHC features linear complexity on both computation and communication with respect to the dimension of set elements. We leverage CKKS, a homomorphic encryption scheme supporting approximate real-valued arithmetic, to compute similarity scores and threshold comparison, along with a clever packing method for efficiency. Moreover, we introduce a novel proof technique to harmonize the approximation error from the sign function with the noise flooding, proving the security of FPHC under the semi-honest model. Finally, we extend our FPHC to functionalities such as labeled or circuit fuzzy PSI. Through experiments, we demonstrate that FPHC can perform fuzzy PSI over 512-dimensional data in a few minutes, which was computationally infeasible in other previous fuzzy PSI proposals.

INDEX TERMS Fuzzy matching, fuzzy private set intersection, homomorphic encryption.

I. INTRODUCTION

In recent years, data-driven technologies such as artificial intelligence (AI) and data collaboration have rapidly advanced, offering innovative and useful services such as biometric identification systems [1], [2], [3], [4], [5], [6], image or text retrieval [7], [8], [9], [10], [11], [12], [13], or recommendation systems [14], [15], [16], [17]. During the operation of these services, one of the essential operations is *similarity matching*, *i.e.*, measuring the similarity between a client's data and a provider's dataset, and allowing the client to learn whether any item in the dataset exceeds a predefined similarity threshold with the client's data. To measure the similarity, the L_1 [18] and L_2 distances [8], and cosine similarity can be used. Among these, cosine similarity has become the dominant similarity metric in the aforementioned

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services [1], [2], [3], [4], [5], [6], [10], [11], [12], [13], [14], [15], [16], [17]. Today, these services are deeply integrated into our lives [11], [13], [17], making the cosine similarity a fundamental component of many real-world applications. However, the exchanged data for similarity matching would contain sensitive information of each party, such as biometrics or location information. It could result in privacy risks such as unauthorized surveillance and potential misuse [19], [20]. Therefore, in compliance with regulations on data privacy such as GDPR [21] or CCPA [22], it is necessary to devise a safeguard for deploying these services.

This motivates the need for systems where the client sends encrypted dataset information to the service provider, who then computes similarity scores using the encrypted data and performs matching based on these scores. Here, all processes must be completed correctly while simultaneously ensuring that no information is revealed except for the final matching result. Particularly, even though the result is derived from the scores, the scores must not be revealed, as it may allow adversaries to infer the information about the data [23], [24], [25]. We describe an example of such a privacy-preserving similarity matching system in Fig. 1.

Private set intersection (PSI) [26], [27], [28], [29], [30] is one of the notable candidates for addressing the challenge of performing the similarity matching while preserving the privacy of datasets. It is a cryptographic protocol that allows parties to compute the intersection between their datasets without revealing any additional information. For decades, several studies have been made to improve the efficiency and functionality of PSI protocols, and these achievements enabled practical deployment of PSI protocols in various application scenarios. For example, Apple and Google have proposed PSI protocols to enable password monitoring [31], collaborative data analysis [32], and the detection of child sexual abuse material (CSAM) [33]. However, traditional PSI protocols have been focused on the *exact* matching only, rather than similarity matching. A naïve approach to support similarity matching is to run the protocols over a set that contains all neighbors of each item held by the service provider within a given similarity threshold in the input space. However, many systems that perform similarity matching deal with high-dimensional data, *e.g.*, 128- to 512-dimensional real-valued vectors in face recognition [1], [2], recommendation systems [14], [16], [17] or speaker recognition [4], [5], [6]. Unfortunately, since the number of neighborhoods is exponential to the dimension of each set element, the approach yields a prohibitive overhead on both computation and communication costs when dealing with high-dimensional data.

Fuzzy private set intersection (fuzzy PSI), which is a PSI protocol supporting both exact and similarity matching, was recently highlighted and actively studied [34], [35], [36], [37], [38], [39], [40], [41]. The fuzzy PSI protocols have been proposed for various similarity metrics, including Hamming distance between binary vectors or Minkowski distances (L_p for $p \in \mathbb{N} \cup \{\infty\}$) between integer-valued vectors. Moreover, several proposals have achieved various functionalities, such as labeled or circuit fuzzy PSI [38], [39], or stronger security notions, including sender privacy [38], [42] or malicious security [37]. Recently, there has been increasing interest in fuzzy PSI protocols for Minkowski distances [38], [40], [41], [42], [43], [44]. Among them, the protocols for the L_2 distance can be used for the cosine similarity; for unit vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ such that $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 = 1$, where $\|\cdot\|_2$ denotes the L_2 norm, we have that $\|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{2 - 2 \cdot \langle \mathbf{x}, \mathbf{y} \rangle}$. Here, $\langle \mathbf{x}, \mathbf{y} \rangle$ corresponds to the cosine similarity between them. However, as discussed in the next section, we found that previous fuzzy PSI protocols require strong assumptions about the data distribution that do not hold for practical scenarios with cosine similarity.

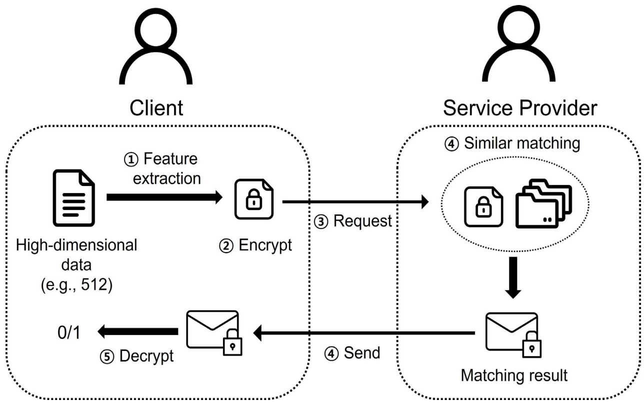

FIGURE 1. The description of similarity matching between a client and a service provider. The service provider learns nothing during the process, while the client learns only a single bit 1 if there exists any service provider's data that exceeds a predetermined similarity threshold with the client's data, and 0 otherwise.

A. LIMITATION OF PREVIOUS PROPOSALS

Previous fuzzy PSI protocols for the L_2 distance share a common protocol design: assign each set item to identifiers and use them as keys to encode key-value pairs. If two parties share common identifiers, they can retrieve the values and obtain the matching result. To avoid collisions between the identifiers, these protocols used several assumptions on the input data distribution. For example, Gao et al. [40] and Zhang et al. [43] separated each vector into d/s consecutive subvectors of size s (with $s = 1$ in [40] and chosen by the parties in [43]) and generated the identifiers by adding vectors $\mathbf{y} \in [-T, T]^s$ to each subvector. On the other hand, other works [38], [41], [42], [44] assumed that the vectors, or together with their neighbors within the threshold, are located in disjoint grid cells of side length $2T$ in the input space to consider the cell indices as the identifiers. Particularly, in [40], [41], [42], and [43], the identifier sets well-constructed under their assumptions have sizes that are not exponential with respect to the dimension of the item. As a result, this leads to the computation/communication costs that are polynomial in the dimension.

In a nutshell, such identifier-based protocols require that neighborhoods of set items in L_∞ of radius T should not overlap each other. However, when set items $\mathbf{x}$ are unit vectors, i.e., $\|\mathbf{x}\|_2 = 1$, we found that such an assumption becomes prohibitively restrictive as the dimension grows. For a precise description, let T_2 and T denote the thresholds for the L_2 distance and cosine similarity, respectively, between vectors in the input space. Given the cosine similarity threshold T, as discussed earlier, we can set the L_2 threshold as $T_2 = \sqrt{2-2T}$ and apply the protocols for the L_2 distance by using T_2. The protocol by van Baarsen and Pu [38] requires that any two d-dimensional elements $\mathbf{x}$ and $\mathbf{y}$ in the receiver's set satisfy $\|\mathbf{x}-\mathbf{y}\|_2 \geq 2T_2(\sqrt{d}+1)$. When $\mathbf{x}$ and $\mathbf{y}$ are the unit vectors, the maximum of $\|\mathbf{x}-\mathbf{y}\|_2$ is 2. It leads to a restriction on the possible threshold: $2T_2(\sqrt{d}+1) \leq 2 \Rightarrow T_2 < \frac{1}{\sqrt{d}+1}$. Since T_2 is $\sqrt{2-2T}$, we finally obtain $T > \frac{2d+1}{2d+2}$. For example, if $d = 512$, then the lower bound becomes ≈ 0.999, which is extremely tighter than the threshold of usual face authentication models [1], [2] that use $T \approx 0.3$.

On the other hand, Gao et al. [40] used the d-separated set assumption. It means that for any two elements, there exists at least one dimension in which their components differ by more than $2T_2$. This means that for every element $\mathbf{x}, \mathbf{y}$, $\|\mathbf{x}-\mathbf{y}\|_\infty > 2T_2$, where $\|\cdot\|_\infty$ denotes the L_∞ distance. Note that $\|\mathbf{x}\|_2 \leq \sqrt{d}\|\mathbf{x}\|_\infty$ for any d-dimensional vector $\mathbf{x}$. Thus, the d-separated set assumption implies that $\|\mathbf{x}-\mathbf{y}\|_2 > 2T_2\sqrt{d}$ in the case where the elements $\mathbf{x}, \mathbf{y}$ have the maximum L_2 distance. For unit vectors, this assumption yields the same bound on the threshold similar to [38]. In addition, we can observe that the *1*-separated set assumption of [42] implies the one used in [40]. Similarly, Zhang et al. [43] assumed the s-separated set. When $s = d$, it is the same as the assumption in [40], which is the weakest assumption of them. When $s = 1$, on the other hand, it is the same as the assumption in [42].

Richardson et al. [41] and Piske et al. [44] used the disjoint hash assumption, which means when the d-dimensional input space is tiled into hypercubes of side length $2T_2$, certain (or all) cells contain at most one point. Equivalently, it implies that there exist elements $\mathbf{x}, \mathbf{y}$ with $\|\mathbf{x}-\mathbf{y}\|_\infty > 2T_2$, imposing a large lower bound on T again. Therefore, all the protocols discussed above are not applicable in practice, and an alternative approach tailored for cosine similarity is needed.

B. CONTRIBUTION

In this paper, we propose a novel fuzzy PSI protocol called FPHC, which is tailored for dealing with high-dimensional data using cosine similarity as a similarity metric. To protect the privacy of input datasets during the protocol, we use the CKKS scheme [45], a fully homomorphic encryption (FHE) scheme that supports approximate arithmetic. That is, we can perform the additions and multiplications on encrypted data and, after decryption, obtain approximate results of the computations. The main ingredients of FPHC are as follows: (1) homomorphic computation for similarity scores between real-valued data over an encryption domain and (2) homomorphic evaluation of an approximated sign function on the result of (1) to obtain a matching result. By leveraging the sign function, the receiver can learn whether there exists a sender's item that matches his query item according to a predefined threshold, without accessing the scores. Our design is implemented by only relying on the arithmetic operations supported by the FHE scheme, from computing similarity scores to applying the sign function for privacy. It eliminates the need for additional privacy-preserving tools such as identifier-based techniques, and we can avoid the strong assumptions in the previous fuzzy PSI protocols.

1) DOUBLY EFFICIENT FUZZY PSI FOR HIGH-DIMENSIONAL DATA WITH COSINE SIMILARITY

To the best of our knowledge, FPHC is the first doubly efficient fuzzy PSI protocol—achieving linear communication and computation costs with respect to the dimension of set elements—that is applicable to our target applications. The asymptotic complexities of FPHC are provided in Tab. 1. Specifically, in the protocol, we adopt a clever plaintext encoding method to avoid the computational overhead of naïvely computing similarity scores in a one-to-one manner over individually encrypted set elements. It allows the similarity scores to be computed in parallel through plaintext-ciphertext multiplications, which are cheaper than ciphertext-ciphertext multiplications. Moreover, thanks to the encoding method, the receiver sends only d ciphertexts to the sender. After that, the sender can obtain all similarity scores with just d plaintext-ciphertext multiplications and subsequently evaluate the sign function once to obtain the matching result. In addition, we do not require any assumption on the input data distribution.

2) SECURITY PROOF TECHNIQUE FOR HANDLING APPROXIMATION ERROR

To ensure that no information is leaked except for the final matching result, we introduce a new security proof technique by incorporating the approximation error, which arises from the computation of an approximated circuit, into the noise flooding framework. Note that in [46], Li et al. showed that when using the CKKS scheme, the ciphertext after circuit evaluation could leak information about private inputs of the circuit because of the inherent noise in the ciphertext. In our setting, it may be regarded as the information about the set held by the sender. A typical solution is to employ the noise flooding technique [47] on the ciphertext being sent to the sender. We found that the approximation error can leak additional information not captured by the inherent ciphertext noise, making the original noise flooding technique insufficient. By extending the technique to address the approximation error and carefully setting the FHE

TABLE 1. Comparison of fuzzy PSI protocols across various metrics. N, M: number of set elements held by receiver and sender, respectively. d: dimension of each set element. T: threshold. s: a parameter chosen by the parties, and $0 \leq s \leq d$. n: ring dimension of FHE. Ctxt**: FHE ciphertext.** Enc**: FHE encryption.** PtCtMul**: plaintext-ciphertext multiplication in FHE.** Sign**: evaluation of *sgn* in FHE. 'disjoint hash' means when d-dimensional input space is tiled into hypercubes of side length $2T$, certain (or all) cells contain at most one element. 's-separated' set assumption means that for every pair of items, there exists at least one dimension in every consecutive s dimensions where the components differ by more than $2T$. '$\geq A$' means the distance (or similarity) of elements is at least A.**

Metric	Protocol	Assumption		Communication	Computation	
		Sender	Receiver		Sender	Receiver
$L_{p\in[1,\infty)}$	[38]	-	$\geq 2T(d^{1/p}+1)$	$O(T^pM + T2^d dN)$	$O((d+T^p)M)$	$O(M + T2^d dN)$
	[40]	d-separated	d-separated	$O((dT + p\log T)M + dTN)$	$O((dT + p\log T)M + N)$	$O(p\log TM + dTN)$
	[43]	s-separated	s-separated	$O(T^s\frac{d}{s}(M+N) + pM\log T)$	$O((T^s\frac{d}{s} + p\log T)M + N)$	$O(T^s\frac{d}{s}N + p\log TM)$
	[42]	$\geq 2Td^{1/p}$	$\geq 2T(d^{1/p}+1)$	$O(N2^d(d+p\log T) + M(dT + p\log T))$	$O(N2^d(d+p\log T) + M(dT + p\log T))$	$O(2^dN(d+p\log T)$
		1-separated	1-separated	$O(N(d+p\log T) + M(dT + p\log T))$	$O(N(d+p\log T) + M(dT + p\log T))$	$O(N(d+p\log T))$
L_2	[41]	disjoint hash	disjoint hash	$O(Nd2^s\log(dT) + M2^{d-s}(\log(dT)d + \log(dT)^3))$	$O((\log(dT^2)d + d + \log(dT^2))^3M2^{d-s})$	$O((\log(dT^2)d + d + \log(dT^2))^3N2^s)$
	[44]	disjoint hash	-	$O(d(N2^d + MT))$	$O(dMT)$	$O(dN2^d + M)$
Cos. Sim.	**Ours**	-	-	$d + \lceil\frac{2MN}{n}\rceil$ Ctxt	$\lceil\frac{2MN}{n}\rceil d$ PtCtMul, $\lceil\frac{2MN}{n}\rceil$ Sign	d Enc

parameters accordingly, we prove that FPHC is secure under semi-honest adversaries.

Labeled/circuit fuzzy PSI: We show that FPHC can be extended to its functionalities without significant overhead, such as *labeled* fuzzy PSI or *circuit* fuzzy PSI, to cover broader application scenarios. For example, in image retrieval, the receiver needs to learn the corresponding label of the intersecting set elements. Note that the matching result of FPHC corresponds to a ciphertext that contains 1 if a match exists, and 0 otherwise. By multiplying a label on the ciphertext, we can easily embed the label. In addition, by cleverly packing the plaintext during the protocol, we show that the arithmetic circuit, such as summation, can be evaluated by applying homomorphic operations on the ciphertext from the sign function evaluation.

3) IMPLEMENTATION

As a proof-of-concept, we instantiate FPHC using the sign function approximation by Cheon et al. [48] and OpenFHE [49] library. FPHC successfully performs the fuzzy PSI on 512-dimensional data in a few minutes, which was computationally infeasible, not even considered, in the previous proposals. To facilitate further study, we released our source code on github.[1]

C. RELATED WORKS

1) TRADITIONAL PSI

Efficient traditional PSI protocols have been proposed in many studies through various cryptographic primitives: oblivious transfer (OT) [50], [51], [52], oblivious pseudorandom function [28], [53], [54], [55], vector oblivious linear evaluation (VOLE) [28], [29], oblivious key-value store (OKVS) [27], [29], [30], [56], and fully homomorphic encryption [26], [57], [58], [59], [60]. Additionally, there are PSI protocols that output related information for the intersection, such as the cardinality [60], [61], [62], [63], [64], the associated labels for each set element [26], [54], [58], [65], or the evaluation of arbitrary circuit [28], [59]. In particular, for FHE-based constructions. Chen et al. [57] proposed the first efficient PSI protocol, along with several optimization tricks. Based on their framework, there has been a series of improvements, including achieving malicious security [58], improving concrete efficiency [26], and extending functionalities [58], [59], [60]. However, all these protocols are designed for exact matching, so they are not applicable in our target scenarios.

2) FUZZY PSI

Fuzzy PSI was first introduced by Freedman et al. [66], and recently several fuzzy PSI protocols have been proposed for various distance metrics. For Hamming distance, Uzun et al. [34] proposed a HE-based construction, and Chakraborti et al. [36] improved it. However, their approaches suffer from non-negligible false-positive rates. For Minkowski distances, which is our main interest, Garimella et al. [35], [37] proposed structure-aware PSI (sa-PSI) for the L_∞ distance, where the receiver only holds a set A of disjoint L_∞ balls. They showed that, under data distribution assumptions like the 1-separated assumption, a fuzzy PSI protocol can be efficiently constructed. They used weak boolean function secret sharing (bFSS) and spatial hashing technique, and their techniques are later improved at [39] to reduce the costs of prior works [35], [37]. On the other hand, van Baarsen and Pu [38] improved the idea of prior works [35], [37] by employing the idea of Apple's PSI [67], relying on a milder assumption. Additionally, they extended their idea to handle the $L_{p\in[1,\infty)}$ distances. Later, van Baarsen and Pu [42] proposed efficient protocols for the L_p based on oblivious

[1] https://github.com/Cryptology-Algorithm-Lab/FPHC

pseudorandom functions (OPRF), including the first protocol for the L_∞ that does not rely on assumptions about the input sets. They achieved greater concrete efficiency than [38]. Gao et al. [40] proposed a fuzzy PSI for the L_p distances based on a novel primitive called fuzzy mapping, which enables them to improve all prior works under a similar assumption. Richardson et al. [41] introduced a fuzzy PSI framework for the L_1, L_2, and L_∞ distances using symmetric key operations, with logarithmic dependency on the distance threshold. Recently, Zhang et al. [43] also proposed a fuzzy PSI protocol for the L_p distances using a refined fuzzy mapping and symmetric key techniques under the s-separated assumption ($0 \leq s \leq d$), where the strength of the assumption depends on s. Piske et al. [44] proposed a new modular framework for fuzzy PSI, and they reduced the design of the protocol based on the L_∞, L_1 and L_2 distances to a novel variant of OT, called distance-aware random OT (da-ROT), reducing the costs of previous works [38], [40], [41]. Although fuzzy PSI protocols for the L_2 distance can be used for dealing with the cosine similarity, as we mentioned, all of them are not applicable in our setting; these proposals have exponential computation and/or communication costs with respect to the dimension or rely on the data distribution assumptions that are inapplicable when considering the cosine similarity.

3) PRIVATE SIMILARITY SCORE COMPUTATION/MATCHING

Along with fuzzy PSI studies, private similarity computation/matching protocols have also been proposed [68], [69], [70], [71], [72], [73], [74], [75], which exhibit somewhat different functionalities in terms of information leakage. For example, a line of work focused on how to compute similarity scores on an encrypted database securely and efficiently. These protocols have built on FHE-based solutions, using several optimization techniques to compute cosine similarity or Euclidean distance in an encrypted domain [68], [70], [71], [74]. Furthermore, for some scenarios where the query privacy is unnecessary, recent studies in secure face recognition tasks further achieved substantial efficiency improvements, thus making them usable in practice [72], [73]. In addition to those works, some studies have focused on reducing the cost for threshold comparison, e.g., group-based threshold testing [76] or a hybrid-style sign function approximation [74].

Note that these protocols cannot be directly compared with fuzzy PSI protocols, as they do not fully satisfy the fuzzy PSI functionality. First, works focused on similarity computation allow the receiver to learn the similarity scores between the sender's items and the receiver's queried ones. Although such information leakage is admissible for some application scenarios, e.g., the trustworthy receiver makes a decision based on scores [70], in general, the leakage of similarity scores exposes several vulnerabilities of the system [23], [25], [77], [78], [79], [80]. Notably, recent studies have demonstrated that the score leakage leads to several privacy attack surfaces, e.g., recovering the sender's item from similarity scores through several carefully designed queries [23], [80]. Putting aside them, some works with secure comparison [73], [74], [76] consider a weaker security notion than fuzzy PSI in a semi-honest model. They consider the scenario where the receiver outsources the encrypted template database to the untrusted 3rd party cloud service, and their main goal is to ensure the receiver's query privacy. For this reason, they consider the scenario with a semi-honest adversarial server without noise flooding or IND-CPA^D capability [46], [47].

II. PRELIMINARIES

Notations: For an integer d, $[d]$ denotes a set of integers $\{1, 2, \ldots, d\}$. We use bold font with uppercase letters to represent matrices, bold font with lowercase letters to represent vectors, and italic font to represent sets. For example, matrix $\mathbf{P}$, vector $\mathbf{c}$, and set I. For a vector $\mathbf{c}$, c_k denotes the k-th component of $\mathbf{c}$. For a matrix $\mathbf{P}$, we use $\mathbf{p}^{(i,\cdot)}$ as the i-th row and $\mathbf{p}^{(\cdot,j)}$ as the j-th column. $p^{(i,j)}$ denotes a component of the matrix $\mathbf{P}$ at position (i, j). For $\mathbf{P} \in \mathbb{R}^{d\times N}$ and $\mathbf{Q} \in \mathbb{R}^{d\times M}$, $\mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)}$ denotes the inner product operation between the i-th column of $\mathbf{P}$ and the j-th column of $\mathbf{Q}$, i.e., $\mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)} = \sum_{k=1}^{d} p^{(k,i)} q^{(k,j)}$. $S^{d-1} = \{\mathbf{c} \in \mathbb{R}^d \mid ||\mathbf{c}||_2 = 1\}$ denotes a unit $(d-1)$-sphere.

For any distribution D, $x \leftarrow D$ denotes sampling x from the distribution D, and it denotes the sampling from the uniform distribution over D when D is a finite set. For a real $\sigma > 0$, $N_{\mathbb{Z}^n}(0, \sigma^2\mathbf{I}_n)$ denotes a discrete multivariate normal distribution with n-dimensional mean vector equal to a zero vector and covariance matrix $\sigma^2\mathbf{I}_n$, where $\mathbf{I}_n$ is an $n \times n$ identity matrix. When $n = 1$, we use $N_{\mathbb{Z}}(0, \sigma^2)$ for simplicity. For an integer q, we identify $\mathbb{Z} \cap (-q/2, q/2]$ as a representative of $\mathbb{Z}_q$. For an integer m, $\mathbb{Z}_m^* = \{x \in \mathbb{Z}_m \mid \gcd(x, m) = 1\}$. $\zeta = \exp(-2\pi i/m)$ denotes a primitive m-th root of unity.

A. FUZZY PRIVATE SET INTERSECTION FOR COSINE SIMILARITY

Our goal is to design a fuzzy PSI protocol for the cosine similarity, which is a two-party protocol between a receiver $\mathcal{R}$ and a sender $\mathcal{S}$. Suppose $\mathcal{R}$ and $\mathcal{S}$ have datasets $\mathbf{P} = \{\mathbf{p}^{(\cdot,1)}, \ldots, \mathbf{p}^{(\cdot,N)}\} \subseteq S^{d-1}$ and $\mathbf{Q} = \{\mathbf{q}^{(\cdot,1)}, \ldots, \mathbf{q}^{(\cdot,M)}\} \subseteq S^{d-1}$, respectively. For a threshold $T \in [-1, 1]$, the goal of the protocol is to let $\mathcal{R}$ learn indices $\{(\mathrm{idx}_1, \mathrm{idx}_2) : \mathbf{p}^{(\cdot,\mathrm{idx}_1)} \cdot \mathbf{q}^{(\cdot,\mathrm{idx}_2)} > T \text{ for } \mathrm{idx}_1 \in [N], \mathrm{idx}_2 \in [M]\}$, without leaking any information to $\mathcal{S}$. We denote a corresponding ideal functionality as $\mathcal{F}_{\mathtt{FPSI}}$, which is described in Fig. 2.

Security Model: We consider a semi-honest adversary [81], *i.e.*, each party honestly follows the protocol, but one of them can be corrupted. We provide the formal definition for semi-honest security. We borrow the notations in [82].

Let us consider a two-party protocol π for computing $\mathcal{F} = (\mathcal{F}_1, \mathcal{F}_2)$, where one party computes $\mathcal{F}_1(x, y)$ on input x and the other computes $\mathcal{F}_2(x, y)$ on input y. Then, π securely

Ideal Functionality $\mathcal{F}_{\texttt{FPSI}}$
Parameters: threshold $T \in [-1, 1]$.
Inputs: $\mathcal{R}$ has input $\mathbf{P} = \{\mathbf{p}^{(\cdot,1)}, \ldots, \mathbf{p}^{(\cdot,N)}\} \subseteq S^{d-1}$. $\mathcal{S}$ has input $\mathbf{Q} = \{\mathbf{q}^{(\cdot,1)}, \ldots, \mathbf{q}^{(\cdot,M)}\} \subseteq S^{d-1}$.
Output: Returns an index set I to $\mathcal{R}$ such that for each $\mathrm{idx}_1 \in [N]$ and $\mathrm{idx}_2 \in [M]$, $(\mathrm{idx}_1, \mathrm{idx}_2) \in I$ if $\mathbf{p}^{(\cdot,\mathrm{idx}_1)} \cdot \mathbf{q}^{(\cdot,\mathrm{idx}_2)} > T$. $\mathcal{S}$ receives nothing.

FIGURE 2. Ideal functionality $\mathcal{F}_{\texttt{FPSI}}$.

computes $\mathcal{F}$ in the presence of static semi-honest adversaries if it satisfies the following definition.

Definition 1 ([82]): Let $\mathcal{F} = (\mathcal{F}_1, \mathcal{F}_2)$ be a functionality. We say that π securely computes a functionality $\mathcal{F}$ in the presence of static semi-honest adversaries if there exist probabilistic polynomial-time algorithms S_1 and S_2 such that

$$\begin{aligned}&\{\mathsf{S}_1(1^n, x, \mathcal{F}_1(x, y)), \mathcal{F}(x, y)\}_{x,y,n}\\ &\overset{c}{\equiv} \{\mathsf{view}_1^{\pi}(x, y, n), \mathsf{output}^{\pi}(x, y, n)\}_{x,y,n}, \ \textit{and}\\ &\{\mathsf{S}_2(1^n, y, \mathcal{F}_2(x, y)), \mathcal{F}(x, y)\}_{x,y,n}\\ &\overset{c}{\equiv} \{\mathsf{view}_2^{\pi}(x, y, n), \mathsf{output}^{\pi}(x, y, n)\}_{x,y,n},\end{aligned}$$

where x and y are inputs of the protocol, and $n \in \mathbb{N}$ is a security parameter.

Here, $\mathsf{view}_i^{\pi}(x, y, n)$ denotes the view of party P_i during an execution of π, including the messages it received. $\mathsf{output}^{\pi}(x, y, n)$ denotes the outputs of the parties. In addition, the notation $\overset{c}{\equiv}$ means computational indistinguishability between two probability ensembles.

Later, we show that the proposed protocol $\pi_{\texttt{FPHE}}$ securely implements the ideal functionality $\mathcal{F}_{\texttt{FPSI}}$ under the semi-honest model: see Theorem 2 for details.

B. CKKS

We use the CKKS scheme [45], an approximate FHE, to perform operations with encrypted real-valued data. As an approximate FHE scheme, the decrypted values obtained by evaluating an arithmetic circuit on encrypted messages approximate the results obtained by directly evaluating the same circuit on the original messages. The scheme is set up with the following parameters: an initial modulus q_0, a scaling factor $\Delta > 0$, a ring dimension n, and the maximum level $L \in \mathbb{N}$. For each level l, the modulus is defined as $q_l = \Delta^l \cdot q_0$ for $0 \le l \le L$. Then, it operates on the following spaces: plaintext space $\mathscr{R} = \mathbb{Z}[X]/\langle X^n + 1\rangle$ and ciphertext space $\mathscr{R}_{q_l} = \mathbb{Z}_{q_l}[X]/\langle X^n + 1\rangle$. Here, n is a power of two, and $X^n + 1$ is a m-th cyclotomic polynomial, where $m = 2n$.

A canonical embedding $\delta : \mathbb{R}[X]/\langle X^n + 1\rangle \to \mathbb{C}^{n/2}$ maps a polynomial $m(X)$ to $n/2$ values among the n evaluations of $m(X)$ at the roots of $X^n + 1$. Specifically, n evaluations are given by $(m(\zeta^k))_{k\in\mathbb{Z}_m^*}$, where ζ is a primitive m-th root of unity. Since $m(\zeta^k) = \overline{m(\zeta^{-k})}$, these n evaluations can be represented as a vector in $\mathbb{C}^{n/2}$. Then, a message $\mathbf{m}$ is encoded to $\lfloor \Delta\delta^{-1}(\mathbf{m})\rceil \in \mathscr{R}$, and $m(X)$ is decoded to $\Delta^{-1}\delta(m(X))$.

Below, we provide a brief description of the scheme, with details available in [45]. Note that a polynomial is sampled from a distribution D by independently sampling each of its coefficients from D.

- $\mathsf{KeyGen}(1^\lambda) \to (pk, evk, sk)$: Given the security parameter λ, set parameters involving m and σ, and output a public key pk, an evaluation key evk, and a secret key sk.
- $\mathsf{Enc}_{pk}(m(X)) \to \mathbf{c} \in \mathscr{R}_{q_L}^2$: Output a ciphertext $\mathbf{c}$ of plaintext $m(X) \in \mathscr{R}$.
- $\mathsf{Dec}_{sk}(\mathbf{c}) \to m'(X)$: For a ciphertext $\mathbf{c}$ of a plaintext $m(X)$, output $m'(X) = m(X) + e(X)$ with an error $e(X)$.
- $\mathsf{Add}(\mathbf{c}_1, \mathbf{c}_2) \to \mathbf{c}_{\mathsf{add}}$: For ciphertexts of $m_1(X)$ and $m_2(X)$, output a ciphertext of $m_1(X) + m_2(X)$, where it has an error that is bounded by the sum of two errors for the inputs.
- $\mathsf{Mult}_{evk}(\mathbf{c}_1, \mathbf{c}_2) \to \mathbf{c}_{\mathsf{mult}}$: For ciphertexts of $m_1(X)$ and $m_2(X)$, output a ciphertext $\mathbf{c}_{\mathsf{mult}}$, where its decryption is $(m_1(X) + e_1(X))(m_2(X) + e_2(X)) + e_{\mathsf{mult}}(X)$ for some polynomial $e_{\mathsf{mult}}(X) \in \mathscr{R}$.
- $\mathsf{RS}_{l\to l'}(\mathbf{c}) \to \mathbf{c}' \in \mathscr{R}_{q_{l'}}^2$: For $\mathbf{c} \in \mathscr{R}_{q_l}^2$, output $\mathbf{c}' \leftarrow \lfloor (q_{l'}/q_l) \cdot \mathbf{c}\rceil \pmod{q_{l'}}$.

The CKKS scheme satisfies IND-CPA security under the hardness assumption of the RLWE problem [83]. Decisional RLWE problem is defined as the computational indistinguishability between the tuple $(as+e, a) \in \mathscr{R}_q^2$, where $s \leftarrow \chi_{\mathsf{Ham}}$ and $e \leftarrow N_{\mathbb{Z}}(0, \sigma^2)$, and a uniformly random tuple $(b, a) \leftarrow \mathscr{R}_q^2$.

C. SIGN FUNCTION

The sign function is one of the fundamental operations, especially for implementing the comparison operator. Due to its discontinuity, homomorphically computing a sign function has been a non-trivial problem, and several methods [48], [84], [85], [86], [87] have been proposed to efficiently compute its approximation. Note that we also need this for determining whether two set elements are sufficiently close or not. First, we define the comparison function and the sign function as follows.

Definition 2 ([48]): A comparison function $comp : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and a sign function $sgn : \mathbb{R} \to \mathbb{R}$ are defined as

$$comp(X, Y) = \begin{cases} 1 & \text{if } X > Y,\\ \frac{1}{2} & \text{if } X = Y,\\ 0 & \text{if } X < Y\end{cases}$$

$$sgn(X) = \begin{cases} 1 & \text{if } X > 0,\\ 0 & \text{if } X = 0,\\ -1 & \text{if } X < 0.\end{cases}$$

Note that $comp(X, Y) = \frac{sgn(X-Y)+1}{2}$.

For an approximated polynomial $p(X)$ of $sgn(X)$, we characterize its precision by defining the notion of (α, ϵ)-close. Each parameter describes how *close* $p(x)$ is to *sgn* and how much $p(x)$ can be faithfully approximated near 0, respectively. The formal definition is given as follows:

Definition 3 ([48]): For $\alpha > 0$ and $0 < \epsilon < 1$, a polynomial $p(X)$ is said to be (α, ϵ)-close to $sgn(X)$ over $[-1, 1]$ if $p(X)$ satisfies the following:

$$\|p(X) - sgn(X)\|_{\infty,[-1,-\epsilon]\cup[\epsilon,1]} \leq 2^{-\alpha},$$

where $||\cdot||_{\infty,R}$ denotes the infinity norm over the domain R. In this paper, we call $|p(x) - sgn(x)|$ an *accuracy error* for $x \in [-1, -\epsilon] \cup [\epsilon, 1]$.

D. NOISE FLOODING

In our construction, the receiver finally obtains a ciphertext corresponding to an output of homomorphic evaluation of some circuit. However, it is known that such a ciphertext in the CKKS scheme could leak information about non-public inputs of the circuit [46], which is the dataset of the sender in our case. Hence, we employ noise flooding to prevent this issue.

The effect of noise flooding in the CKKS scheme was thoroughly studied by Li et al. [47]. For the sake of a rigorous security proof in our protocol, we introduce some results of theirs. Their analysis was based on the previous result [46] that the leakage of non-public inputs stems from the decryption error, which is defined as follows:

Definition 4 ([47]): Let $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec}, \mathsf{Add}, \mathsf{Mult}, \mathsf{RS})$ be a (approximation) FHE scheme with plaintext space $\mathcal{M} \subseteq \widetilde{\mathcal{M}}$, which is a normed space with norm $\|\cdot\| : \widetilde{\mathcal{M}} \to \mathbb{R}_{\geq 0}$. For any ciphertext $\mathbf{c}$, plaintext m, and secret key sk, the ciphertext error of $(\mathbf{c}, m, sk)$ is defined to be

$$\mathsf{Error}(\mathbf{c}, m, sk) = \|\mathsf{Dec}_{sk}(\mathbf{c}) - m\|.$$

For the CKKS scheme, canonical embedding norm is used, which is defined by for $a(X) \in \mathbb{Q}[X]/\langle X^n + 1\rangle$, $\|a(X)\|_\infty^{\mathsf{can}} = \|(a(\zeta^k))_{k\in\mathbb{Z}_m^*}\|_\infty$.

Suppose $\mathbf{c}_1, \ldots, \mathbf{c}_k$ are encryptions of plaintexts $m_1, \ldots, m_k$, respectively. Given an arithmetic circuit $\mathcal{G}$, let $\mathbf{c}$ denote a ciphertext computed on $\mathbf{c}_1, \ldots, \mathbf{c}_k$ for $\mathcal{G}$, and let m be an output of $\mathcal{G}(m_1, \ldots, m_k)$. In this case, we call the ciphertext error $\mathsf{Error}(\mathbf{c}, m, sk)$ a *circuit error*. Li et al. [85] proved that one can blend the non-trivial information from the circuit error via a similar scale of the noise, hence achieving the desirable goal. More precisely, one can achieve q-IND-CPA^D security with this modification, which means that the adversary cannot obtain any information about the non-public inputs with up to q queries to a decryption oracle. We provide the full statement as follows:

Definition 5 ([47]): Let ct be a tuple $(\mathsf{ct}.c, \mathsf{ct}.t)$ for the CKKS scheme, where $\mathsf{ct}.c$ is a ciphertext, and $\mathsf{ct}.t$ is an estimation of the worst error bound for $\mathsf{ct}.c$. Then, for $\sigma^* > 0$, S-CKKS$_{\sigma^*}$ is the CKKS scheme that modifies decryption to compute S-CKKS$_{\sigma^*}.\mathsf{Dec}_{sk}(\mathsf{ct}) = \mathsf{Dec}_{sk}(\mathsf{ct}.c) + e$ for $e \leftarrow N_{\mathbb{Z}^n}(0, \sigma^{*2}\mathsf{ct}.t^2\mathbf{I}_n)$.

Theorem 1 ([47]): For any $q, n \in \mathbb{N}$, let q be the maximum of decryption queries an attacker can make, and let n be the ring dimension. Then, if CKKS is $(c + \log_2 24)$-bit IND-CPA-secure, and $\sigma^* = \sqrt{24qn}2^{s/2}$, then S-CKKS$_{\sigma^*}$ is (c, s)-bit q-IND-CPA^D-secure.

Here, s is a parameter for statistical security, *i.e.*, the adversary can gain any non-trivial information from a decrypted result with probability 2^{-s}. We also note that the worst error bound can be estimated according to [45].

In this work, we use the idea of adding noise so that the receiver cannot obtain any information about the sender's set from the final ciphertext with up to q decryptions.

III. FUZZY PSI PROTOCOL FOR COSINE SIMILARITY

We now present our Fuzzy PSI protocol for cosine similarity, which is specialized for dealing with high-dimensional data. The protocol is denoted by $\pi_{\mathtt{FPHC}}$. At a high level, it consists of the following four steps. We represent the receiver and the sender as $\mathcal{R}$ and $\mathcal{S}$, respectively.

1) $\mathcal{R}$ encrypts its data via the CKKS scheme and sends it to $\mathcal{S}$.
2) $\mathcal{S}$ homomorphically computes the cosine similarity between each data.
3) $\mathcal{S}$ homomorphically evaluates a comparison function with the threshold.
4) $\mathcal{S}$ applies the noise flooding technique to the result and sends it to $\mathcal{R}$.

In the following subsections, we provide details about each step, along with an analysis of the computational and communication costs. We also prove its security in the semi-honest model.

A. CONSTRUCTION OF PROTOCOL FPHC

1) SETTINGS AND DATA FORMAT

We denote $\mathbf{P} = \{\mathbf{p}^{(\cdot,1)}, \ldots, \mathbf{p}^{(\cdot,N)}\} \subseteq S^{d-1}$ and $\mathbf{Q} = \{\mathbf{q}^{(\cdot,1)} \ldots, \mathbf{q}^{(\cdot,N)}\} \subseteq S^{d-1}$ as datasets held by $\mathcal{R}$ and $\mathcal{S}$, respectively. Note that $\mathcal{R}$ and $\mathcal{S}$ can normalize their elements to lie on S^{d-1}. For the ease of explanation, we assume that $MN = n/2$ for the ring dimension n, *i.e.*, MN is equal to the number of packed components in one CKKS ciphertext. When $MN > n/2$, we run the protocol $\lceil \frac{2MN}{n} \rceil$ times by splitting either $\mathbf{P}$ or $\mathbf{Q}$ into smaller subsets and executing the protocol between each subset and the other party's dataset. Later, we provide more details in Section III-B. We assume that $\mathbf{P}$ and $\mathbf{Q}$ are well-quantized so that for sufficiently small $\epsilon > 0$, the inner product between each pair of set elements does not lie within the range $(T - 2\epsilon, T + 2\epsilon)$. Note that this is not a harsh assumption because reducing precision, *e.g.*, 8-bit or 16-bit quantization methods, has been shown not to significantly affect inner product values [88], [89]. Throughout this paper, we use $\epsilon = 2^{-16}$.

2) SETUP

Each dataset can be represented as matrices, *i.e.*, $\mathbf{P} \in \mathbb{R}^{d\times N}$ and $\mathbf{Q} \in \mathbb{R}^{d\times M}$, respectively. In this step, $\mathcal{R}$ generates the

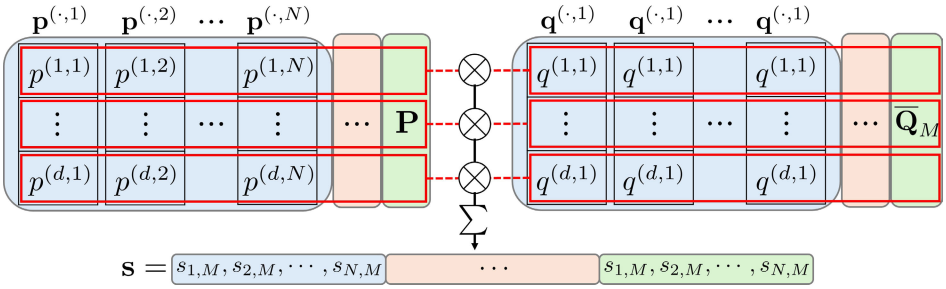

FIGURE 3. The description of our approach to compute cosine similarity scores. $\otimes$ and Σ denote component-wise multiplication and summation, respectively.

key pair (pk, evk, sk), and $\mathcal{R}$ and $\mathcal{S}$ encode their datasets into plaintexts. Our key idea is to cleverly encode $\mathbf{P}$ and $\mathbf{Q}$ to enable the computation of similarity scores using homomorphic operations between plaintext and ciphertext, which are much cheaper than ciphertexts. This process is described in Figure 3. First, $\mathcal{R}$ sets $\overline{\mathbf{P}}$ by concatenating M copies of $\mathbf{P}$. For $j \in [M]$, $\mathcal{S}$ creates $\overline{\mathbf{Q}}_j$, each containing N copies of $\mathbf{q}^{(\cdot,j)}$. After that, $\mathcal{S}$ sets $\overline{\mathbf{Q}}$ by concatenating $\overline{\mathbf{Q}}_j$'s. Finally, they each encode the rows $\overline{\mathbf{p}}^{(i,\cdot)}$ of $\overline{\mathbf{P}}$ and $\overline{\mathbf{q}}^{(i,\cdot)}$ of $\overline{\mathbf{Q}}$ for $i \in [d]$.

3) COMPUTATION OF SIMILARITY SCORE

In this step, the goal is to homomorphically compute $\mathbf{P}^T \cdot \mathbf{Q} \in \mathbb{R}^{N\times M}$. The plaintext-ciphertext multiplication between the i-th rows of $\overline{\mathbf{P}}$ and $\overline{\mathbf{Q}}$ corresponds to element-wise multiplication. As shown in Figure 3, summing the results $\overline{\mathbf{v}}_i$ obtained from only d multiplications generates a single ciphertext that contains all scores $s_{i,j} = \mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)}$ between $\mathbf{P}$ and $\mathbf{Q}$. This single ciphertext is $\mathbf{c}$, and it allows the desired result to be computed with a single evaluation of the comparison function.

4) EVALUATING COMPARISON FUNCTION

After computing the cosine similarity scores, $\mathcal{S}$ runs a comparison circuit with respect to the threshold T, *i.e.*, $\frac{sgn(\mathbf{w})+1}{2}$. To this end, $\mathcal{S}$ utilizes a polynomial $p(X)$ that is $(\alpha - 1, \epsilon)$-close to the sign function sgn over the interval $[-1, 1]$. Here, one caveat is that the value of $\mathbf{c} - T$ lies within $[-2, 2]$ because $\mathbf{c}$ is the ciphertext of the scores (i.e., values in $[-1, 1]$) and T is also in $[-1, 1]$. Thus, we need to homomorphically multiply $\frac{1}{2}$ with $\mathbf{c} - T$ to fit the desired range of $[-1, 1]$, and the result is $\mathbf{w}$. Additionally, note that $(\alpha - 1, \epsilon)$-closeness does not guarantee that the evaluation of $p(X)$ on the interval $(-\epsilon, \epsilon)$ approximates the sign function. Therefore, we must ensure that the value of $\mathbf{w}$ does not lie within $(-\epsilon, \epsilon)$. In other words, the value of $\mathbf{c}$ must not fall within the interval $(T - 2\epsilon, T + 2\epsilon)$. As mentioned above, we can address this problem using the quantization. After evaluation, we can ensure that the resulting ciphertext $\mathbf{s}$ holds a message that lies within $[1 - 2^{-\alpha}, 1 + 2^{-\alpha}]$ if there is an overlap between set elements, otherwise lies within $[-2^{-\alpha}, 2^{-\alpha}]$.

5) NOISE FLOODING TECHNIQUE

To ensure the privacy of $\mathcal{S}$'s dataset, $\mathcal{S}$ finally applies the noise flooding technique on $\mathbf{s}$. Recall that Theorem 1 tells us the precise amount of the noise for whitening the information leakage from the circuit error. However, in our setting, this theorem *as is* would not ensure the security of our protocol because of the approximation error on evaluating *sgn*. Hence, instead of adding the noise according to the worst-case circuit error $\mathsf{ct}.t$, we need to use an adjusted value $\overline{\mathsf{ct}.t}$ that also considers the approximation error.

We also note that the decrypted values are also affected by the noise flooding. For this reason, $\mathcal{R}$ employs a parameter r to determine the overlapping indices from the received ciphertext $\mathbf{s}$. More precisely, for the message $\mathbf{z}$ corresponding to $\mathbf{s}$, $\mathcal{R}$ checks whether each component of $\mathbf{z}$ lies within $[1 - 2^r, 1 + 2^r]$ or $[-2^r, 2^r]$. The precise analysis on the desired $\overline{\mathsf{ct}.t}$ and r will be provided in the next subsection, along with the security proof of our protocol.

6) THE FULL PROTOCOL

By merging these building blocks all together, we finally construct our Fuzzy PSI protocol $\pi_{\mathtt{FPHC}}$. After executing the protocol, $\mathcal{R}$ can derive the result from the message $\mathbf{z}$. Due to the encoding method, the message contains approximations of 1 or 0 corresponding to all MN similarity scores; starting from the N scores between $\mathcal{S}$'s $\mathbf{q}^{(\cdot,1)}$ and all vectors of $\mathcal{R}$, and continuing sequentially up to the N scores between $\mathcal{S}$'s $\mathbf{q}^{(\cdot,M)}$ and all vectors of $\mathcal{R}$. Therefore, for $\mathrm{idx}_1 \in [N]$ and $\mathrm{idx}_2 \in [M]$, $\mathcal{R}$ checks whether each component $z_{(\mathrm{idx}_2-1)N+\mathrm{idx}_1}$ is an approximation of 1 or not. If so, $\mathcal{R}$ concludes that $\mathbf{p}^{(\cdot,\mathrm{idx}_1)} \cdot \mathbf{q}^{(\cdot,\mathrm{idx}_2)} > T$. For completeness, we describe the overall protocol with details in Fig. 4.

B. EFFICIENCY ANALYSIS

We now analyze the computation and communication cost of $\pi_{\mathtt{FPHC}}$ for each party. We first consider the case when $MN = n/2$ and extend it to the general case. During the protocol, the communication occurs when sending and retrieving ciphertexts, *i.e.*, d ciphertexts for encrypting $\overline{\mathbf{P}}$ and a ciphertext corresponding to $\mathbf{s}$. That is, the communication cost is $(d + 1)$ ciphertexts. For computation, we first observe that $\mathcal{R}$ proceeds with d encryption and 1 decryption

Parameters: threshold $T \in [-1, 1]$, and security parameters λ and r.
Inputs: $\mathcal{R}$ has input $\mathbf{P} = \{\mathbf{p}^{(\cdot,1)}, \ldots, \mathbf{p}^{(\cdot,N)}\} \subseteq S^{d-1}$. $\mathcal{S}$ has input $\mathbf{Q} = \{\mathbf{q}^{(\cdot,1)}, \ldots, \mathbf{q}^{(\cdot,M)}\} \subseteq S^{d-1}$. i.e., $\mathbf{P} \in \mathbb{R}^{d\times N}$ and $\mathbf{Q} \in \mathbb{R}^{d\times M}$.
Setup:

1) $\mathcal{R}$ runs $(pk, evk, sk) \leftarrow \mathsf{KeyGen}(1^\lambda)$.
2) $\mathcal{R}$ sets a matrix $\overline{\mathbf{P}} = \mathbf{P}\| \cdots \|\mathbf{P} \in \mathbb{R}^{d\times n/2}$ by concatenating M copies of $\mathbf{P}$.
3) For each $i \in [d]$, $\mathcal{R}$ encodes $\overline{\mathbf{p}}^{(i,\cdot)}$ to a polynomial $\overline{p}^{(i,\cdot)}(X) \in \mathscr{R}$.
4) For $j \in [M]$, $\mathcal{S}$ creates $\overline{\mathbf{Q}}_j \in \mathbb{R}^{d\times N}$ by concatenating N copies of $\mathbf{q}^{(\cdot,j)}$, then sets $\overline{\mathbf{Q}} = \overline{\mathbf{Q}}_1\| \cdots \|\overline{\mathbf{Q}}_M \in \mathbb{R}^{d\times n/2}$.
5) For each $i \in [d]$, $\mathcal{S}$ encodes $\overline{\mathbf{q}}^{(i,\cdot)}$ to a polynomial $\overline{q}^{(i,\cdot)}(X) \in \mathscr{R}$.

Procedure:

1) For each $i \in [d]$, $\mathcal{R}$ runs $\mathbf{p}_i \leftarrow \mathsf{Enc}_{pk}(\overline{p}^{(i,\cdot)}(X))$.
2) $\mathcal{R}$ sends pk, evk, and $\{\mathbf{p}_i\}_{i\in[d]}$ to $\mathcal{S}$.
3) For each $i \in [d]$, $\mathcal{S}$ computes $\mathbf{v}_i = \overline{q}^{(i,\cdot)}(X) \cdot \mathbf{p}_i$ and $\overline{\mathbf{v}}_i \leftarrow \mathsf{RS}(\mathbf{v}_i)$.
4) $\mathcal{S}$ computes $\mathbf{c}$ by adding $\overline{\mathbf{v}}_i$ for all $i \in [d]$ using the Add algorithm.
5) $\mathcal{S}$ computes $\overline{T} \leftarrow \lfloor \Delta T \rceil \in \mathscr{R}$ and $\mathbf{b} = \mathbf{c} - (\overline{T}, 0)$.
6) $\mathcal{S}$ computes $a \leftarrow \lfloor \Delta \frac{1}{2} \rceil \in \mathscr{R}$, $\overline{\mathbf{w}} = a \cdot \mathbf{b}$, and $\mathbf{w} \leftarrow \mathsf{RS}(\overline{\mathbf{w}})$.
7) $\mathcal{S}$ computes $\mathbf{s}$ for the sign function sgn with $\mathbf{w}$ by using the Add, Mult and RS algorithms.
8) $\mathcal{S}$ samples $e \leftarrow N_{\mathbb{Z}^n}(0, {\sigma^*}^2 \overline{\mathsf{ct}.t}^2 \mathbf{I_n})$ and computes $\overline{\mathbf{z}} = \mathbf{s} + (e, 0)$.
9) $\mathcal{S}$ sends $\overline{\mathbf{z}}$ to $\mathcal{R}$.
10) $\mathcal{R}$ runs $z(X) \leftarrow \mathsf{Dec}_{sk}(\overline{\mathbf{z}})$ and decodes $z(X)$ to obtain a message $\mathbf{z}$.
11) For $\mathbf{z} = (z_1, \ldots, z_{n/2})$, $\mathcal{R}$ sets an index set I satisfying $(\mathrm{idx}_1, \mathrm{idx}_2) \in I \subseteq [N] \times [M]$ if and only if $z_{(\mathrm{idx}_2-1)N+\mathrm{idx}_1} \in [1 - 2^r, 1 + 2^r]$.

FIGURE 4. The description of the proposed fuzzy PSI protocol $\pi_{\mathtt{FPHC}}$.

during the protocol. On the other hand, $\mathcal{S}$ takes $(d + 1)$ plaintext-ciphertext multiplications and 1 evaluation of the sign function.

We now consider the case $MN > n/2$. If we assume $N < n/2$, then we can chunk the dataset $\mathbf{Q}$ of $\mathcal{S}$ into $k = \lceil \frac{2MN}{n} \rceil$ pieces, say $\mathbf{Q}_1, \ldots, \mathbf{Q}_k$, where each piece has elements at most $\frac{n}{2N}$. Hence, it suffices to run k runs of $\pi_{\mathtt{FPHC}}$ with inputs $(\mathbf{P}, \mathbf{Q}_1), \ldots, (\mathbf{P}, \mathbf{Q}_k)$ for $\mathcal{R}$ and $\mathcal{S}$, respectively. Here, we can observe that for each protocol, $\mathcal{R}$ sends the same ciphertexts to $\mathcal{S}$. Hence, the computation cost of $\mathcal{R}$ remains the same as $\pi_{\mathtt{FPHC}}$ and the communication cost only increases by k, whereas the computation of $\mathcal{S}$ is scaled by k.

C. SECURITY ANALYSIS

In this subsection, we show that the proposed $\pi_{\mathtt{FPHC}}$ indeed achieves the semi-honest security, *i.e.*, securely implements the ideal functionality $\mathcal{F}_{\mathtt{FPSI}}$ in Fig. 2. The core part of our security proof is to analyze the scale $\overline{\mathsf{ct}.t}$ of noise flooding to ensure that the retrieved ciphertext $\mathbf{s}$ did not contain any useful information. For this reason, we will focus on providing our analysis about $\overline{\mathsf{ct}.t}$.

1) DERIVING $\overline{\mathsf{CT}.T}$ FOR NOISE FLOODING

We first recall that the rationale of employing the noise flooding is to cover the circuit error via a large noise. For a precise description, let us denote $x_0 = \mathsf{Dec}_{sk}(\mathsf{ct}.c) = m(X) + e_c(X)$ as a decryption result with a circuit error $e_c(X)$ and $x_1 = m(X)$. Then the effect of the noise flooding can be viewed as adding $e^*(X)$ on both x_0 and x_1, ensuring that $e^*(X)$ and $e_c(X) + e^*(X)$ are statistically indistinguishable. The Theorem 1 tells us the scale of noise when $\|e_c(X)\|_\infty^{\mathsf{can}} \le \mathsf{ct}.t$.

On the other hand, in our setting, one more thing needs to be considered: if we decrypt $\mathbf{s}$, say $s(X) = \mathsf{Dec}_{sk}(\mathbf{s})$, then $s(X)$ contains not only the circuit error but also the accuracy error. More precisely, if we denote acc_i as the accuracy error of the i'th component, *i.e.*, the difference between the evaluation result of $p(X)$ and the actual comparison function, then we have that $\mathsf{acc}_i \in [-2^{-\alpha}, 2^{-\alpha}]$ for $i \in [n/2]$. Note that $\mathsf{acc}_i \in \mathbb{R}$, so in the context of plaintexts, we need to consider an encoded polynomial $e_{\mathsf{acc}}(X) = \lfloor \Delta\delta^{-1}((\mathsf{acc}_i)_{i\in[n/2]}) \rceil$. This can be interpreted as an additional error added to the circuit error $e_c(X)$. That is, by using the same argument as above and Theorem 1, it suffices to compute the upper bound of $\|e_c(X)+e_{\mathsf{acc}}(X)\|_\infty^{\mathsf{can}}$ for applying the noise flooding. Note that by the property of canonical norm and the triangular inequality, we obtain that

$$\begin{aligned} ||e_c(X) + e_{\mathsf{acc}}(X)||_\infty^{\mathsf{can}} &\le ||e_c(X)||_\infty^{\mathsf{can}} + ||e_{\mathsf{acc}}(X)||_\infty^{\mathsf{can}} \\ &\le \mathsf{ct}.t + \Delta 2^{-\alpha}. \end{aligned}$$

Therefore, for the scale term $\overline{\mathsf{ct}.t} = \mathsf{ct}.t + \Delta 2^{-\alpha}$, we finally can exploit the result of Theorem 1 by replacing $\mathsf{ct}.t$ with $\overline{\mathsf{ct}.t}$.

2) PARAMETER SELECTION

To ensure the correctness of our protocol $\pi_{\mathtt{FPHC}}$, $\mathcal{R}$ should be able to distinguish whether the retrieved ciphertext from $\mathcal{S}$ with noise flooding contains 0 or 1. That is, two intervals

$[1-2^r, 1+2^r]$ and $[-2^r, 2^r]$ in the previous section should be non-overlapping. In this paragraph, we show that this is possible by a careful parameter selection of the CKKS scheme. Therefore, combining with Theorem 1, we can conclude that $\pi_{\texttt{FPHC}}$ securely implements the ideal functionality $\mathcal{F}_{\texttt{FPSI}}$.

To address this, let us assume that the noise $e^*(X)$ from $N(0, \sigma^{*2}\overline{\mathsf{ct}.t}^2\mathbf{I}_n)$ is added to $\mathbf{s}$, in accordance with Theorem 1. Note that $\sigma^* = \sqrt{24qn}2^{s/2}$ for parameters $q, s \in \mathbb{N}$, which will be chosen later. Then we can observe that the total error compared to the result from the actual comparison function becomes

$$\|e_c(X) + e_{acc}(X) + e^*(X)\|_\infty^{\mathsf{can}} \le \overline{\mathsf{ct}.t} + \|e^*(X)\|_\infty^{\mathsf{can}}.$$

Hence, the error corresponding to each message is at most $\frac{\overline{\mathsf{ct}.t}+\|e^*(X)\|_\infty^{\mathsf{can}}}{\Delta}$.

To analyze this, we attempt to put $\beta\sqrt{n}\sigma^*\overline{\mathsf{ct}.t}$ in place of $\|e^*(X)\|_\infty^{\mathsf{can}}$, where β is a parameter determined later. Then we obtain

$$\begin{aligned}\frac{\overline{\mathsf{ct}.t} + \beta\sqrt{n}\sigma^*\overline{\mathsf{ct}.t}}{\Delta} &= (\beta\sqrt{n}\sigma^* + 1)\left(\frac{\mathsf{ct}.t}{\Delta} + 2^{-\alpha}\right)\\ &= (1 + \beta n\sqrt{24q}2^{s/2})\left(\frac{\mathsf{ct}.t}{\Delta} + 2^{-\alpha}\right).\end{aligned} \tag{1}$$

If we choose $q = 2^5$, $s = 40$, and $n = 2^{17}$, then the R.H.S. of the above equality becomes $\approx \beta \cdot 2^{41.79}(\frac{\mathsf{ct}.t}{\Delta} + 2^{-\alpha})$.

Here, note that the random variable $e^*(\zeta^k)$ for $k \in \mathbb{Z}_m^*$ follows $N_{\mathbb{Z}}(0, n\sigma^{*2}\overline{\mathsf{ct}.t}^2)$, since it is a linear combination of n independent samples of $N_{\mathbb{Z}}(0, \sigma^{*2}\overline{\mathsf{ct}.t}^2)$ with weights $\{\zeta^{k(i-1)}\}_{i=1}^n$. Since all random variables $\{e^*(\zeta^k)\}_{k\in\mathbb{Z}_{m^*}}$ are mutually independent because of its structure, we obtain

$$\Pr[\|e^*(X)\|_\infty^{\mathsf{can}} > \beta\sqrt{n}\sigma^*\overline{\mathsf{ct}.t}] \tag{2}$$
$$= 1 - \left(1 - \Pr[|e^*(\zeta^k)| > \beta\sqrt{n}\sigma^*\overline{\mathsf{ct}.t}]\right)^n$$
$$< np \tag{3}$$

for any $k \in \mathbb{Z}_m^*$, where $p = \Pr[|e^*(\zeta^k)| > \beta\sqrt{n}\sigma^*\overline{\mathsf{ct}.t}]$. The last inequality holds because $(1-p)^n \ge 1 - np$. Finally, by using the fact that $\Pr_{Z\sim N_{\mathbb{Z}(0,1)}}[|Z| > \lambda] \le e^{-\frac{\lambda^2}{2}}$ [90], we obtain $\Pr[\|e^*(X)\|_\infty^{\mathsf{can}} > \beta\sqrt{n}\sigma^*\overline{\mathsf{ct}.t}] < n \cdot e^{-\frac{\beta^2}{2}} < n \cdot 2^{-\frac{\beta^2}{2}}$.

If we select $\beta = 16 = 2^4$, then the above bound becomes 2^{-111}. Therefore, by setting Δ and α to satisfy $2^{-1} > 2^4 \cdot 2^{41.79}(\frac{\mathsf{ct}.t}{\Delta} + 2^{-\alpha})$ and $\Delta > 2^5 \cdot 2^{41.79}(\mathsf{ct}.t + \Delta \cdot 2^{-\alpha})$, we can expect that $\mathcal{R}$ successfully achieves its goal with an error of probability 2^{-111}. Note that by adding the noise, we can ensure that $\mathcal{R}$ can not obtain the information about the sender's set with up to q decryptions. Here, q becomes the number of possible runs of our protocol without regenerating keys. That is, after 2^5 runs of $\pi_{\texttt{FPHC}}$ without changing the key, both $\mathcal{R}$ and $\mathcal{S}$ regenerate the key pair.

Under the above parameter setting, we finally show the semi-honest security of the $\pi_{\texttt{FPHC}}$ protocol, as stated in Theorem 2.

Theorem 2: The protocol $\pi_{\texttt{FPHC}}$ described in Figure 4 securely implements the ideal functionality $\mathcal{F}_{\texttt{FPSI}}$ under the static semi-honest model.

Proof: Due to the parameter settings described earlier, the index set and the output of the functionality are identical with a negligible failure probability. Hence, it suffices to show the existence of probabilistic polynomial-time algorithms S_1 and S_2, which simulate the views of the receiver and the sender, respectively.

First, we show how S_1 simulates the receiver's view in the real protocol. Specifically, the receiver's view consists of $\overline{\mathbf{z}} \in \mathscr{R}^2$. The simulator S_1 receives the index set I as the output of the functionality. From the set, it generates a vector $\mathbf{z}' = (z'_1, \ldots, z'_{n/2})$ as follows.; If $(i,j) \in I$ for $i \in [N], j \in [M]$, the simulator sets $z'_{(j-1)N+i} = 1$. Similarly, if $(i,j) \notin I$, it sets $z'_{(j-1)N+i} = 0$. Then, S_1 runs $\overline{\mathbf{z}}' \leftarrow \mathsf{Enc}_{pk}(\mathbf{z}')$ and add $(e, 0)$ to it for $e \leftarrow N_{\mathbb{Z}^n}(0, \sigma^{*2}\overline{\mathsf{ct}.t}^2\mathbf{I}_n)$.

The remaining part is to show the indistinguishability between $\overline{\mathbf{z}}$ and $\overline{\mathbf{z}}'$ with the knowledge of secret key. To this end, we consider four hybrids $\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2$ and $\mathcal{H}_3$ as follows. For simplicity, let $\mathcal{N}$ denote $N_{\mathbb{Z}^n}(0, \sigma^{*2}\overline{\mathsf{ct}.t}^2\mathbf{I}_n)$.

Hybrid $\mathcal{H}_0$.

$$(\overline{\mathbf{z}},\ \mathsf{Dec}_{sk}(\overline{\mathbf{z}})) = (\overline{\mathbf{z}},\ m + e + e_0^*),\ \ e_0^* \leftarrow \mathcal{N}$$

Here, e denotes the error of $\overline{\mathbf{z}}$ before applying the noise flooding, and m is a plaintext except for the error. $\mathcal{H}_0$ corresponds to the view of the real execution.

Hybrid $\mathcal{H}_1$. We now consider $\mathcal{H}_1$, which removes e from the decryption output of $\overline{\mathbf{z}}$ in $\mathcal{H}_0$.

$$(\overline{\mathbf{z}},\ m + e_1^*),\ \ e_1^* \leftarrow \mathcal{N}$$

Hybrid $\mathcal{H}_2$. In this hybrid, the decryption output of $\overline{\mathbf{z}}$ in $\mathcal{H}_0$ is replaced by that of $\overline{\mathbf{z}}'$. Since S_1 sets the vector $\mathbf{z}'$ so that the plaintexts of $\overline{\mathbf{z}}$ and $\overline{\mathbf{z}}'$ are identical except for their errors, $\mathcal{H}_2$ is defined as

$$(\overline{\mathbf{z}},\ \mathsf{Dec}_{sk}(\overline{\mathbf{z}}')) = (\overline{\mathbf{z}},\ m + e_{\mathsf{fresh}} + e_2^*),\ \ e_2^* \leftarrow \mathcal{N}$$

Here, e_{fresh} exactly corresponds to the error introduced during encryption, and both e and e_{fresh} have canonical embedding norms bounded by $\overline{\mathsf{ct}.t}$.

Hybrid $\mathcal{H}_3$. Lastly, we replace $\overline{\mathbf{z}}$ in $\mathcal{H}_2$ with $\overline{\mathbf{z}}'$.

$$(\overline{\mathbf{z}}',\ \mathsf{Dec}_{sk}(\overline{\mathbf{z}}'))$$

$\mathcal{H}_3$ corresponds to the view simulated by S_1.

Now, we demonstrate that $\mathcal{H}_0$ and $\mathcal{H}_3$ are computationally indistinguishable.

1) Indistinguishability of $\mathcal{H}_0$ and $\mathcal{H}_1$. The difference between the two hybrids is the random variable of the right component: $\mathcal{H}_0$ contains $e + e_0^*$, whereas $\mathcal{H}_1$ contains e_1^*. As shown in Section III-C, for any e bounded by $\overline{\mathsf{ct.t}}$ and noises e_0^*, e_1^* sampled from $\mathcal{N}$, the distributions of $e + e_0^*$

and e_1^* are statistically indistinguishable. Therefore, $\mathcal{H}_0$ and $\mathcal{H}_1$ are also statistically indistinguishable.

2) Indistinguishability of $\mathcal{H}_1$ and $\mathcal{H}_2$. By the same argument, the distributions of e_1^* and $e_{\mathsf{fresh}} + e_2^*$ are also statistically indistinguishable. Thus, $\mathcal{H}_1$ and $\mathcal{H}_2$ are indistinguishable.

3) Indistinguishability of $\mathcal{H}_2$ and $\mathcal{H}_3$. The only difference between these two hybrids is the ciphertext in the left component: $\mathcal{H}_2$ includes $\overline{\mathbf{z}}$, while $\mathcal{H}_3$ includes $\overline{\mathbf{z}}'$. Since the CKKS scheme is IND-CPA secure under the hardness assumption of the Ring-LWE problem, the ciphertexts $\overline{\mathbf{z}}$ and $\overline{\mathbf{z}}'$ are computationally indistinguishable.

From 1)–3), we conclude that $\mathcal{H}_0$ and $\mathcal{H}_3$ are computationally indistinguishable. Consequently, the receiver's view in the real protocol and the simulated view are computationally indistinguishable.

Finally, we show how S_2 simulates the sender's view in the real protocol. Specifically, the sender's view consists of $pk, evk,$ and $\{\mathbf{p}_i\}_{i\in[d]}$. Similarly, the simulator S_2 can simulate the sender's view by uniformly sampling $2d + 4$ components from $\mathscr{R}$. By the IND-CPA security of CKKS under the Ring-LWE assumption, the sender's view and simulated view are computationally indistinguishable. □

IV. EXTENDING FUNCTIONALITY OF PROTOCOL FPHC

As we can figure out in Fig. 2, the π_{FPHC} is only designed for the scenario where the receiver is sufficient to learn whether the input elements are overlapping with some other elements of the sender or not. However, this functionality is insufficient for covering other practical applications. For example, in privacy-preserving biometric search [34], the receiver needs the label of the matching item, while in contact tracing [91], the number of intersection is required. For this reason, we now focus on extending the functionality of π_{FPHC}.

A. LABELED AND CIRCUIT FUZZY PSI

For the candidate of extended functionalities, we consider the labeled fuzzy PSI and circuit fuzzy PSI, which are analogues of labeled PSI [26], [54], [58] and circuit PSI [28], [59] for traditional PSIs. For the labeled fuzzy PSI, we assume that there is a label $l_j \in \mathcal{L} \subset \mathbb{N}$ assigned to each set element $\mathbf{q}^{(\cdot,j)}$ for $j \in [M]$ held by the sender $\mathcal{S}$. We assume that these labels are represented by w-bit (unsigned) integers, so for all $l \in \mathcal{L}$, $l < 2^w$ holds. From this setting, $\mathcal{R}$ wishes to learn pairs of indices $(i, j) \in [N] \times [M]$ and a label l_j such that $\mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)} > T$. For indices of non-overlapping set elements, $\mathcal{R}$ gets $0 \in \mathbb{Z}$.

On the other hand, in the circuit fuzzy PSI, $\mathcal{R}$ wishes to learn the evaluation result of an arbitrary function f that takes N integers $v_1, \ldots, v_N$ such that $v_i = |\{j \in [M] : \mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)} > T\}|$ as inputs. We denote ideal functionalities of labeled fuzzy PSI and circuit fuzzy PSI as $\mathcal{F}_{\mathtt{LFPSI}}$ and $\mathcal{F}_{\mathtt{CFPSI}}$, and we describe the functionalities in Fig. 6 and Fig. 7.

Achieving the labeled fuzzy PSI is straightforward in our construction. Specifically, $\mathcal{S}$ sets a label vector

$$\mathbf{l} = (\underbrace{l_1, \ldots, l_1}_{N \text{ copies}}, \ldots, \underbrace{l_M, \ldots, l_M}_{N \text{ copies}}) \in (\{0,1\}^w)^{MN},$$

which consists of N copies of each label l_j, where each l_j is the label of $\mathbf{q}^{(\cdot,j)}$ for $j \in [M]$. Note that we set $MN = n/2$. Also, $\mathcal{S}$ generates a polynomial $\ell(X)$ by encoding the label vector $\mathbf{l}$. Let $c_{i,j} = comp(\mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)}, T)$. Then, with ignoring errors, the result of the homomorphic evaluation for the comparison function $\mathbf{s}$ contains a decoded message $\mathbf{z}$ denoted by

$$\mathbf{z} = (c_{1,1}, \ldots, c_{N,1}, c_{1,2}, \ldots, c_{N,2}, \ldots, c_{N,M}) \in \mathbb{R}^{MN}.$$

If the sender performs a plaintext-ciphertext multiplication between $\ell(X)$ and $\mathbf{s}$, the receiver obtains pairs (i, j) and labels l_j when $\mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)} > T$. We denote the labeled fuzzy PSI protocol constructed in this manner as $\pi_{\mathtt{LFPHC}}$.

The security proof also remains the same, except for a simple treatment on analyzing an additional noise accompanied by the label. We provide a proof in Theorem 3.

Theorem 3: The protocol $\pi_{\mathtt{LFPHC}}$ securely implements the ideal functionality $\mathcal{F}_{\mathtt{LFPSI}}$ under the static semi-honest model.

Proof: We focus on the analysis of parameter selection and the simulation of the receiver's view, as the remaining parts of the proof are the same as in Theorem 2.

Let $\mathbf{s}_{\text{label}}$ be a result of the plaintext-ciphertext multiplication between $\ell(X)$ and $\mathbf{s}$. Since the labels are w-bit unsigned integers, the accuracy error in the i'th component of a message decoded from $\mathsf{Dec}_{sk}(\mathbf{s}_{\text{label}})$ lies within $[-2^{w-\alpha}, 2^{w-\alpha}]$. Thus, we must set the parameter $\overline{\mathsf{ct}.t} = \mathsf{ct}.t + \Delta 2^{w-\alpha}$. For $e^* \leftarrow N_{\mathbb{Z}^n}(0, \sigma^{*2}\overline{\mathsf{ct}.t}^2\mathbf{I}_n)$, the sender sends $\overline{\mathbf{z}}_{\text{label}} = \mathbf{s}_{\text{label}} + (e^*, 0)$ to the receiver as the final result. Here, $\sigma^* = \sqrt{24qn}2^{s/2}$, where q and s are parameters defined in Theorem 1, and n is the ring dimension. As discussed in Section III-C, the total error in each message of $\overline{\mathbf{z}}_{\text{label}}$ is at most $\frac{\overline{\mathsf{ct}.t}+\|e^*(X)\|_\infty^{\mathsf{can}}}{\Delta} = (1 + \beta n\sqrt{24q}2^{s/2})\left(\frac{\mathsf{ct}.t}{\Delta} + 2^{w-\alpha}\right)$. Hence, if we set $q = 2^5$, $s = 40$, $n = 2^{17}$, and $\beta = 2^4$, we must choose the parameters Δ and α such that $2^{-1} > 2^4 \cdot 2^{41.79}\left(\frac{\mathsf{ct}.t}{\Delta} + 2^{w-\alpha}\right)$.

A simulator S, which simulates the receiver's view, pairs (i, j) and labels l_j as the output of functionality $\mathcal{F}_{\mathtt{LFPSI}}$ if $\mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)} > T$. As in Theorem 2, S generates a vector $\mathbf{z}' = (z_1', \ldots, z_{n/2}')$ as follows: for the given pairs (i, j), it sets $z'_{(j-1)N+i} = l_j$, while setting all other components to 0. As a result, S encrypts $\mathbf{z}'$ and applies the noise flooding to simulate the result that the sender sends to the receiver. □

1) LEVERAGING THE PACKING STRUCTURE FOR CIRCUIT FUZZY PSI

Achieving the circuit fuzzy PSI seems to be rather tricky compared to the labeled fuzzy PSI. Nevertheless, we show that our $\pi_{\mathtt{FPHC}}$ can be easily extended to achieve this functionality by utilizing the structure of packed messages. Assume $\mathbf{z} = (c_{1,1}, \ldots, c_{N,1}, c_{1,2}, \ldots, c_{N,2}, \ldots, c_{N,M})$.

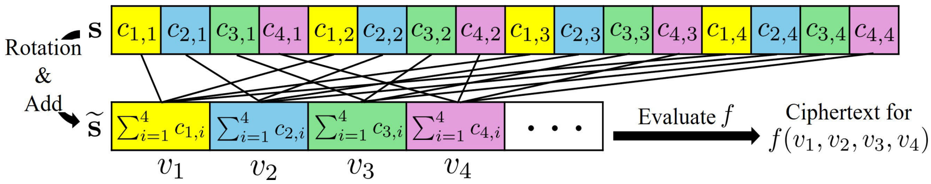

FIGURE 5. The description of our approach to compute cosine similarity scores. ⊗ and Σ denote component-wise multiplication and summation, respectively.

Ideal Functionality $\mathcal{F}_{\texttt{LFPSI}}$
Parameters: threshold $T \in [-1, 1]$.
Inputs: $\mathcal{R}$ has input $\mathbf{P} = \{\mathbf{p}^{(\cdot,1)}, \ldots, \mathbf{p}^{(\cdot,N)}\} \subseteq S^{d-1}$. $\mathcal{S}$ has inputs $\mathbf{Q} = \{\mathbf{q}^{(\cdot,1)}, \ldots, \mathbf{q}^{(\cdot,M)}\} \subseteq S^{d-1}$ and. $\mathcal{L} = \{l_j \in \{0,1\}^w : j \in [M]\}$.
Output: Returns pairs of indices $(i,j) \in [N] \times [M]$ and labels l_j such that $\mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)} > T$

FIGURE 6. Ideal functionality $\mathcal{F}_{\texttt{LFPSI}}$.

Ideal Functionality $\mathcal{F}_{\texttt{CFPSI}}$
Parameters: threshold $T \in [-1, 1]$, and an arbitrary function f.
Inputs: $\mathcal{R}$ has input $\mathbf{P} = \{\mathbf{p}^{(\cdot,1)}, \ldots, \mathbf{p}^{(\cdot,N)}\} \subseteq S^{d-1}$. $\mathcal{S}$ has inputs $\mathbf{Q} = \{\mathbf{q}^{(\cdot,1)}, \ldots, \mathbf{q}^{(\cdot,M)}\} \subseteq S^{d-1}$.
Output: Returns the evaluation result of an arbitrary function f that takes N integers $v_1, \ldots, v_N$ such that $v_i = |\{j \in [M] : \mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)} > T\}|$ as inputs.

FIGURE 7. Ideal functionality $\mathcal{F}_{\texttt{CFPSI}}$.

We can observe that the subvector $\mathbf{z}_i := (\mathbf{z}[i + jN])_{j=0}^{M-1}$ of stride N, where $\mathbf{z}[k]$ denotes the k'th component of $\mathbf{z}$, corresponds to $(c_{i,1}, \ldots, c_{i,M})$, *i.e.*, values with the same first index i.

We now explain why this packing structure enables us to achieve the circuit Fuzzy PSI. Because of the structure, we can observe that when rotating $\mathbf{z}$ by N on the left, then the corresponding $\mathbf{z}_i$ becomes $(c_{i,2}, \ldots, c_{i,M}, c_{i,1})$, *i.e.*, rotation on the original subvector $(c_{i,1}, \ldots, c_{i,M})$ by 1 on the left. Therefore, by a folklore rotation-and-add technique, we can compute the ciphertext $\tilde{\mathbf{s}}$ containing M concatenations of $(\sum_{j=1}^{M} c_{1,j}, \sum_{j=1}^{M} c_{2,j}, \ldots, \sum_{i=1}^{M} c_{N,j})$. To help understand, we depict an overview of our idea with the structure of $\mathbf{z}$ in Fig. 5. Since $\sum_{j=1}^{M} c_{i,j} = |\{j \in [M] : \mathbf{p}^{(\cdot,i)} \cdot \mathbf{q}^{(\cdot,j)} > T\}| = v_i$ holds by definition, we finally implement the desired functionality by homomorphically evaluating the circuit f on $\tilde{\mathbf{s}}$. Specifically, let $I(X)$ be an encoding result of a vector of size $n/2$ whose first component is 1 and all remaining components are 0. For each $t \in \{0, \ldots, N-1\}$, by rotating $\tilde{\mathbf{s}}$ to the left by t positions and performing the plaintext-ciphertext multiplication with $I(X)$, we can obtain a ciphertext $\tilde{\mathbf{s}}_{t+1}$ corresponding to $(v_{t+1}, 0, \ldots, 0)$. Then, we can get the output of the circuit f by using $\tilde{\mathbf{s}}_{t+1}$'s.

The security proof goes similarly to Theorem 2. The only difference comes from calculating the scale of noise flooding. To this end, we need to consider the error that appears in the rotation-and-add technique and evaluate f, tracking how much the accuracy error acc is magnified during these operations.

Theorem 4: The protocol $\pi_{\texttt{CFPHC}}$ securely implements the ideal functionality $\mathcal{F}_{\texttt{CFPSI}}$ under the static semi-honest model.

Proof: We focus on analyzing how the accuracy error can be traced as it is magnified during the operations. The remaining part of the proof is the same in Theorem 2 and 3.

To generate each ciphertext $\tilde{\mathbf{s}}_i$, the sender performs the following operations: 1) M rotations and 2) M additions to compute the ciphertext $\tilde{\mathbf{s}}$; 3) N rotations and 4) N plaintext-ciphertext multiplications with the plaintext $I(X)$. In the CKKS scheme, each rotation introduces an additional error that accumulates into the circuit error. However, this does not affect the bound of the accuracy error, since accuracy error is not part of the circuit error. Moreover, since the plaintext used in step 4) is the polynomial $I(X)$, only the first component of the decoded message from each $\tilde{\mathbf{s}}_i$ contains a nonzero value. As a result, after steps 1) to 4), the first component of the message encoded in $\tilde{\mathbf{s}}_i$ contains the value v_i along with a magnified accuracy error, denoted by $\mathsf{acc}_i \in [-M2^{-\alpha}, M2^{\alpha}]$. Note that the accuracy bound is scaled by a factor of M because of M additions in step 2).

Now, we explain how the accuracy error is affected by each homomorphic evaluation. This allows us to track the accuracy error bound when evaluating an arbitrary circuit f. To get $v_1 \cdot v_2$, we can perform the multiplication between the ciphertexts $\tilde{\mathbf{s}}_1$ and $\tilde{\mathbf{s}}_2$. Suppose each $\tilde{\mathbf{s}}_i$ corresponds to an encrypted message of the form $(v_i + \mathsf{acc}_i, 0, \ldots, 0)$. Then, the first component of the resulting message after multiplication has an accuracy error given by $(v_1 + \mathsf{acc}_1)(v_2 + \mathsf{acc}_2) - v_1 v_2$, which can be bounded using the fact that $v_i \in \{0, 1, \ldots, M\}$. If performing the addition with $\tilde{\mathbf{s}}_1$ and $\tilde{\mathbf{s}}_2$, the accuracy error is additive: $\mathsf{acc}_1 + \mathsf{acc}_2$. Additionally, consider a plaintext $w(X)$ encoding a vector whose components are bounded by B. When performing a plaintext-ciphertext multiplication between $w(X)$ and $\tilde{\mathbf{s}}_1$, the resulting accuracy error is scaled by B, i.e., $B \cdot \mathsf{acc}_1$. By applying these rules recursively, we can track the overall accuracy error bound when evaluating the circuit f on inputs $\tilde{\mathbf{s}}_1, \ldots, \tilde{\mathbf{s}}_N$. However, note that to track the error bound introduced by ciphertext-ciphertext multiplication, we must also keep track of the bound on each computation result excluding the accuracy error. □

V. IMPLEMENTATION RESULTS

We now provide the implementation results of $\pi_{\texttt{FPHC}}$. We used the official Python wrapper of OpenFHE v1.3.1.0 [49] for the CKKS scheme. All experiments were done on a machine with AMD EPYC 7543P CPU (32 cores with 64 threads; 2.8 GHz) and 512 GB of RAM. We utilized all available CPU cores and threads in our implementations.

Details on Instantiation: To implement an approximation sign function that is $(\alpha - 1, \epsilon)$-close to $sgn(X)$, we use the method proposed by Cheon et al. [48]. Specifically, we set $\alpha = 66$ to ensure that the receiver achieves its goal with negligible error probability, and choose $\epsilon = 2^{-16}$. For this, we evaluate a polynomial composed of g_4 (7 times) and f_4 (3 times) by following Algorithm 3 in [48]. To estimate $\mathsf{ct}.t$, we use the static noise estimation in OpenFHE. With $MN = n/2$, we measure $\mathsf{ct}.t \approx 2^{8.1}$; hence, we set $\mathsf{ct}.t = 2^9$ for the noise flooding.

From these settings, the CKKS parameters are chosen as follows: the ring dimension $n = 2^{17}$, the scaling factor $\Delta = 2^{59}$, and the maximum level $L = 41$. The remaining parameters are selected to ensure 128-bit security following the homomorphic encryption standard [92]. In addition, according to our analysis in Section III-C, we apply the noise flooding technique to ensure 40-bit statistical security with $q = 2^5$.

1) COSTS OF PROTOCOL FPHC

To conduct the experiment, we first sampled data from a standard normal distribution and normalized it to lie on the unit hypersphere S^{d-1}. For the performance evaluation, we measured the elapsed time and communication cost for various values of M, increasing M from $2^{11} \approx 2,000$ to $2^{11} \times 5 \approx 10,000$, with fixed d and N. The results in Fig. 8 show that as M increases, the sender's computational cost grows linearly while the receiver's computation cost remains unchanged. The communication cost increases linearly, with only a slight rise from 9.839 GB to 9.848 GB. We note that in one of our target applications, face recognition, state-of-the-art models typically use feature vectors of up to 512 dimensions, as exemplified by ArcFace [1] and AdaFace [2]. Meanwhile, other applications, such as speaker recognition, image retrieval, and recommendation systems, use embeddings with dimensions up to 1024 [4], [6], [7], [11], [14], [17]. In particular, a prior work [93] has suggested that these feature vectors can be further compressed while preserving their discriminative capability. Motivated by these observations, we measured the effect of the dimensionality d by varying d from 2^4 to 2^{10}, while fixing N and M, as shown in Fig. 9. Note that previous studies [38], [40], [41], [43], [44] only considered cases where d is at most 10. As shown in Tab. 2, the computation cost of each operation increases linearly with the dimension, except for FHE parameter/key generation via KeyGen and the computation of the sign

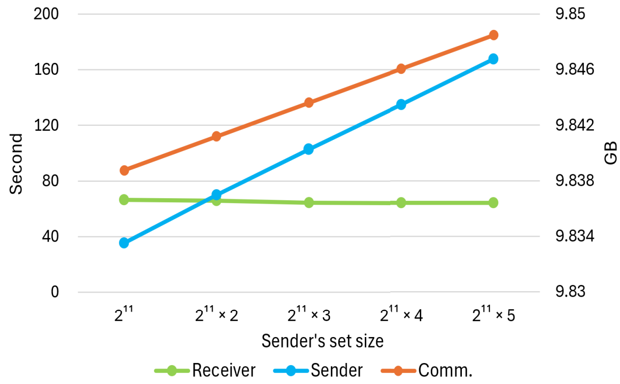

FIGURE 8. Runtime performance and communication cost of the $\pi_{\texttt{FPHC}}$ implementation under varying M with fixed $d = 2^7$.

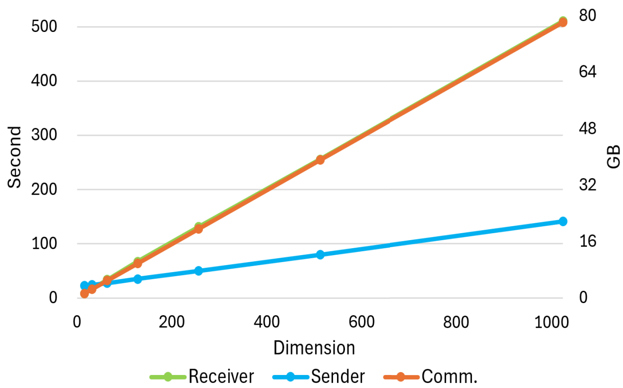

FIGURE 9. Runtime performance and communication cost of the $\pi_{\texttt{FPHC}}$ implementation under varying d with fixed $(N, M) = (2^5, 2^{11})$.

function on the ciphertext space. Both of them remain constant regardless of dimension. Together with Tab. 2, Fig. 9 shows that both the communication and computation costs increase asymptotically linearly with the dimension. We also highlight that $\pi_{\texttt{FPHC}}$ covers the case $d = 2^9$, which was computationally infeasible in all existing works but is necessary to handle real-world applications [1], [2], [6], [7], [14].

2) REAL-WORLD APPLICATIONS

Now, we describe how quantization is applied and then investigate the effect of quantization in real-world applications. Since our protocol employs an (α, ϵ)-close sign function, Theorem 2 guarantees that, when the similarity scores should not be within $(T - 2\epsilon, T + 2\epsilon)$ for a threshold T, the output of the ideal functionality $\mathcal{F}_{\texttt{FPSI}}$ in Figure 2 is equal to the output of our protocol $\pi_{\texttt{FPHC}}$. To satisfy the above condition, we apply quantization, which slightly modifies the original feature vectors.

a: QUANTIZATION

Specifically, before conducting experiments on the real-world datasets, we quantized each vector $\mathbf{x}$ as

$$Q_b(\mathbf{x}) = \lfloor 2^b \cdot \mathbf{x} \rfloor \cdot 2^{-b}.$$

TABLE 2. Cost breakdown by operation of π_{FPHC} under varying d (with fixed $(N, M) = (2^5, 2^{11})$). Gen.para.: time to generate parameters for CKKS and KeyGen. Enc.: time to encrypt the receiver set. Sign: time to evaluate the sign function over ciphertext space. Gen.pt.: time to generate plaintexts for sender set. Inn.prd.: time to evaluate the inner product over ciphertext space.

d	$\mathcal{R}$		$\mathcal{S}$		
	Gen.para.(s)	Enc.(s)	Sign(s)	Gen.pt.(s)	Inn.prod.(s)
2^4	3.809	8.481	19.410	1.125	2.055
2^5		17.013		2.252	4.031
2^6		33.992		4.535	7.755
2^7		67.342		8.992	15.568
2^8		131.077		18.262	30.784
2^9		255.788		36.280	61.035
2^{10}		510.146		73.651	122.514

That is, only b bits in the fractional part are preserved, and the result $Q_b(\mathbf{x})$ can be written as $k \cdot 2^{-b}$ for some integer k. In our setting, $\epsilon = 2^{-16}$, and therefore we selected $b = 7$ so that the similarity scores (i.e., inner products of vectors) are represented as multiples of 2^{-14}. The threshold values also were quantized in the same way. However, in this case, the similarity scores can be equivalent to the quantized threshold. To address this issue, we added a small value of 2^{-15} to the quantized threshold, which guarantees the assumption.

To assess the effect of quantization on real-world applications, we measured the difference in accuracy before and after quantization for the identification and verification protocols. To extract template vectors, we utilized two pre-trained models: ArcFace [1], which employs a ResNet100 backbone and was trained on Glint360K [94], and AdaFace [2], which employs the ViT-KPRPE backbone [95] and was trained on WebFace12M [96]. These models are available from InsightFace [2] and CVLFace ,[3] respectively.

b: IDENTIFICATION PROTOCOL

For the identification protocol, we measured the accuracy before and after quantization on the IJB-C dataset [97] using the 1:N mixed identification protocol. In our experiment, we used three template sets: Gallery 1 (G1), consisting of 1,772 subjects and 1,772 templates; Gallery 2 (G2), consisting of 1,759 subjects and 1,759 templates; and Mixed, consisting of 3,531 subjects and 19,593 templates. Both G1 and G2 include one template per subject, and their subject identities are disjoint. The template extraction process followed the procedure described in [97]. To evaluate the accuracy of the identification, we queried a gallery set with a probe set and computed two accuracy metrics: False Positive Identification Rate (FPIR) and True Positive Identification Rate (TPIR). FPIR is defined as the proportion of probe templates whose subjects are not present in the gallery but whose cosine similarity with at least one gallery template exceeds the threshold. TPIR is defined as the proportion of probe templates whose subjects are present in the gallery and whose cosine similarity with the same subject's template in the gallery exceeds the threshold and is the highest among all gallery templates. To assess the effect of quantization on TPIR, we selected thresholds corresponding to FPIR values of 0.1, 0.01, 0.001, and 0 by using both the original and quantized templates. The TPIR values computed from the original and quantized vectors for each FPIR are shown in Tab. 3. The results show that the quantization only leads to TPIR variations of less than one percentage point for the identification protocol, except for AdaFace in the G1 gallery at FPIR $= 0$, where the difference is less than three percentage points.

[2] https://github.com/deepinsight/insightface
[3] https://github.com/mk-minchul/CVLface

TABLE 3. Threshold and TPIR for original templates and those quantized to 7-bit fractional part, corresponding to model, gallery, and FPIR.

Model	Gallery	FPIR	Threshold		TPIR	
			Original	Quantization	Original	Quantization
ArcFace	G1	0.1	0.268	0.232	0.9750	0.9750
		0.01	0.349	0.302	0.9638	0.9637
		0.001	0.571	0.486	0.8703	0.8776
		0	0.798	0.690	0.4484	0.4496
	G2	0.1	0.270	0.234	0.9587	0.9589
		0.01	0.349	0.301	0.9432	0.9437
		0.001	0.455	0.392	0.9089	0.9094
		0	0.554	0.478	0.8458	0.8458
AdaFace	G1	0.1	0.285	0.247	0.9825	0.9825
		0.01	0.364	0.316	0.9721	0.9722
		0.001	0.553	0.483	0.8983	0.8937
		0	0.794	0.681	0.4371	0.4587
	G2	0.1	0.282	0.243	0.9679	0.9681
		0.01	0.352	0.305	0.9572	0.9569
		0.001	0.477	0.415	0.9121	0.9110
		0	0.624	0.543	0.7802	0.7744

c: VERIFICATION PROTOCOL

For the verification protocol, we measured the accuracy before and after quantization on three datasets—LFW [98], CFP-FP [99], and AgeDB [100]—using their verification protocols. These datasets include matching and non-matching pairs, with 3,000/3,000 pairs in LFW, 3,500/3,500 pairs in CFP-FP, and 3,000/3,000 pairs in AgeDB. Using these pairs, we evaluated two accuracy metrics: False Acceptance Rate (FAR) and True Acceptance Rate (TAR). FAR is defined as the false positive rate over the non-matching pairs, while TAR is defined as the true positive rate over the matching pairs. To assess the effect of quantization, we computed thresholds corresponding to FAR values 0.01, 0.001, and 0, as in the identification protocol, and calculated the corresponding TAR values, shown in Tab. 4. The results show that the quantization only leads to variations in TAR of less than 0.3 percentage points for each verification protocol.

Furthermore, the previous proposals [38], [40], [41], [42], [43], [44] required strong assumptions on input data distributions, which resulted in thresholds as high as 0.999, as mentioned in Section I-A. This makes it difficult to

TABLE 4. Threshold and TAR for original templates and those quantized to 7-bit fractional part, corresponding to model, dataset, and FAR.

Model	dataset	FAR	Threshold		TAR	
			Original	Quantization	Original	Quantization
ArcFace	LFW	0.01	0.145	0.125	0.998	0.998
		0.001	0.191	0.169	0.997	0.997
		0	0.242	0.207	0.997	0.997
	CFP-FP	0.01	0.145	0.124	0.9897	0.9897
		0.001	0.207	0.180	0.9871	0.9871
		0	0.236	0.198	0.9849	0.9857
	AgeDB	0.01	0.180	0.156	0.9753	0.974
		0.001	0.239	0.204	0.9603	0.96
		0	0.265	0.230	0.9533	0.9523
AdaFace	LFW	0.01	0.138	0.121	0.9977	0.9977
		0.001	0.198	0.162	0.9967	0.9967
		0	0.226	0.194	0.9967	0.9967
	CFP-FP	0.01	0.139	0.121	0.9889	0.9889
		0.001	0.191	0.164	0.9869	0.9871
		0	0.211	0.179	0.9851	0.9854
	AgeDB	0.01	0.172	0.148	0.971	0.971
		0.001	0.241	0.206	0.952	0.9543
		0	0.244	0.214	0.951	0.951

set practical threshold values, such as those presented in Tab. 3 and Tab. 4. In contrast, our approach requires only quantization and a small change of the threshold without any assumptions. Thus, FPHC is more suitable for applications in our target scenarios.

VI. DISCUSSION

A. SIGN FUNCTION APPROXIMATION AND ITS IMPACT

Our protocol consumes one level in Step 3 of Fig. 4, while the remaining 40 levels are used for evaluating the sign function. This indicates that the multiplicative depth required for the sign function is the major contributor to the overall depth. Since the modulus size of the ciphertext space grows linearly with the maximum level, this directly impacts the communication cost. Likewise, the computational cost is influenced by both the number of Mult_{evk} operations and the maximum level, as each Mult_{evk} involves ciphertext-domain multiplications. Therefore, the sign function evaluation constitutes the main performance bottleneck of our protocol. If a lower-depth approximation of the sign function with fewer multiplication operations were to be used, both the communication and computation costs could be significantly reduced. Another related issue stems from the approximation interval. The approximation proposed by Cheon et al. [48] has intervals near zero where bounded approximation error cannot be guaranteed. To mitigate this, we applied a well-known quantization technique, which introduces a small accuracy difference. If an approximation guaranteeing the bound of error over the entire domain were adopted, the quantization would not be necessary.

B. MALICIOUS SECURITY

Finally, we note that our construction is analyzed only in the semi-honest security model, where all parties are assumed to follow the protocol honestly. Unfortunately, it does not capture adversaries that may deviate from the protocol or manipulate intermediate values. To achieve malicious security, the protocol would require additional procedures that allow each party to prove that his messages are consistent and that he carries out the protocol honestly, without revealing any private information [39], [101], [102]. However, it may incur computation and communication overhead, making it challenging to preserve the protocol's efficiency.

VII. CONCLUSION

In this work, we propose FPHC, a novel doubly efficient fuzzy PSI protocol designed for handling high-dimensional data with cosine similarity as a similarity metric. FPHC achieves linear communication and computation costs with respect to the dimension, requiring much weaker assumptions compared to several previous Fuzzy PSI proposals [38], [40], [41], [42], [43], [44]. This overcomes the limitations of previous works that either rely on strong assumptions or incur exponential costs. We also show that our FPHC is secure under the semi-honest security model by carefully addressing the information leakage from approximation operations with the noise flooding technique. Additionally, we extend FPHC to support labeled and circuit fuzzy PSI, enriching its applicability to various scenarios.

We leave several interesting future directions. First, we expect that our protocol can be improved by employing more efficient sign function evaluation methods [85], [86], [87]. Of independent interest, it would also be interesting to devise an alternative approach for designing fuzzy PSI protocols without the sign function. In addition, our protocol is based on the semi-honest security model. Extending our protocol to defend against stronger adversaries, such as malicious security, would be another interesting research direction. Finally, with a slight modification, we expect that our framework would support the fuzzy PSI with L_p distance over the Euclidean space, hence increasing its flexibility for a wider range of applications. We leave investigations for these aspects as future work.

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HYUNJUNG SON received the B.S. degree in mathematics from Sookmyung Women's University, Seoul, Republic of Korea, in 2023. She is currently pursuing the Ph.D. degree in applied mathematics with Hanyang University, Seoul. Her research interests include cryptography, especially zero-knowledge proof and private set operations.

SEUNGHUN PAIK (Graduate Student Member, IEEE) received the B.S. degree in mathematics from Hanyang University, Seoul, Republic of Korea, in 2023, where he is currently pursuing the Ph.D. degree in applied mathematics. From 2024 to 2025, he was a Visiting Researcher with the University of Notre Dame. His research interests include applied cryptography, privacy-preserving machine learning, and biometric template protection.

YUNKI KIM received the B.S. degree in mathematics from Hanyang University, Seoul, Republic of Korea, in 2025, where he is currently pursuing the M.S. degree in applied mathematics. His research interests include cryptography, especially private set intersection.

SUNPILL KIM (Graduate Student Member, IEEE) received the B.S. degree in mathematics from Hanyang University, Seoul, Republic of Korea, in 2020, where he is currently pursuing the Ph.D. degree in applied mathematics. From 2023 to 2024, he was a Ph.D. Student Researcher with the Institute for Infocomm Research (I^2R) of the Agency for Science, Technology and Research (A^*STAR) under the Ministry of Trade and Industry of Singapore. His research interests include AI security, deep learning, and cryptography. He has been receiving the First Graduate Presidential Science Scholarship since 2024.

HEEWON CHUNG received the B.S. degree in mathematics from Korea Advanced Institute of Science and Technology (KAIST), Daejon, Republic of Korea, in 2009, and the M.S. and Ph.D. degrees in mathematics from Seoul National University, Seoul, Republic of Korea, in 2017. He was a Cryptographic Researcher with DESILO Inc., Republic of Korea, from 2022 to 2025. He has been an Assistant Professor with the Department of Software Engineering, Jeonbuk National University, Jeonju, Republic of Korea, since 2025. His research interests include primarily all aspects of cryptography, including but not limited to private set operations, the foundations of blockchain, and practical applications using fully homomorphic encryption. He is also interested in solving scalability problems in the blockchain field using zero-knowledge proofs (SNARKs, incrementally verifiable computation, and vector commitment).

JAE HONG SEO received the B.S. degree in mathematics from Korea University, Seoul, Republic of Korea, in 2004, and the Ph.D. degree in mathematics from Seoul National University, Seoul, in 2011. He was with the National Institute of Communications and Technology, Japan, from 2011 to 2013. He was with Myongji University, Republic of Korea, from 2013 to 2018. He has been a Professor with the Mathematics Department, Hanyang University, Seoul, since 2018. His research interests include computational algorithms, cryptology, and their applications in deep learning and blockchain. He served as a Program Committee Member for IACR conferences and workshops. He received the Sangsan Young Mathematician Award from Korean Mathematical Society in 2012 and the Young Scientist Award for Excellence from Elsevier and NRF in 2018. He was the Program Committee Co-Chair of APKC 2018 and ICISC 2019.